

Species Selection to Minimize Biomass Variability

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To reduce variability in aggregated fish biomass data for boosted regression tree modeling, we applied a multi-criteria selection framework using data from 2025 only (the year used for model training). Our objective was to identify the most stable and representative species within each taxonomic family while maintaining ecological realism and ensuring all families were adequately represented.

1 Exploratory Visualization

2 Data Preparation

2.1 Download Data

2.2 Construct species lookup table

Obtain taxonomic information from all species using the rcv package.

https://jeremiaheb.github.io/NCRMP_R_Fish_Package/

```
## SPECIES_CD FAMILY SCINAME COMNAME LC LM
## 1 ABL HIAN Belonidae Ablennes hians flat needlefish NA NA
## 2 ABU SAXA Pomacentridae Abudefduf saxatilis sergeant major NA NA
## 3 ABU TAUR Pomacentridae Abudefduf taurus night sergeant NA NA
## 4 ACA ASPE Chaenopsidae Acanthemblemaria aspera roughhead blenny NA NA
## 5 ACA CHAP Chaenopsidae Acanthemblemaria chaplini papillose blenny NA NA
## 6 ACA MARI Chaenopsidae Acanthemblemaria maria secretary blenny NA NA
## WLEN_A WLEN_B
## 1 NA NA
## 2 0.0000164 3.142
## 3 NA NA
## 4 0.0000084 2.962
## 5 0.0000084 2.962
## 6 0.0000084 2.962
```

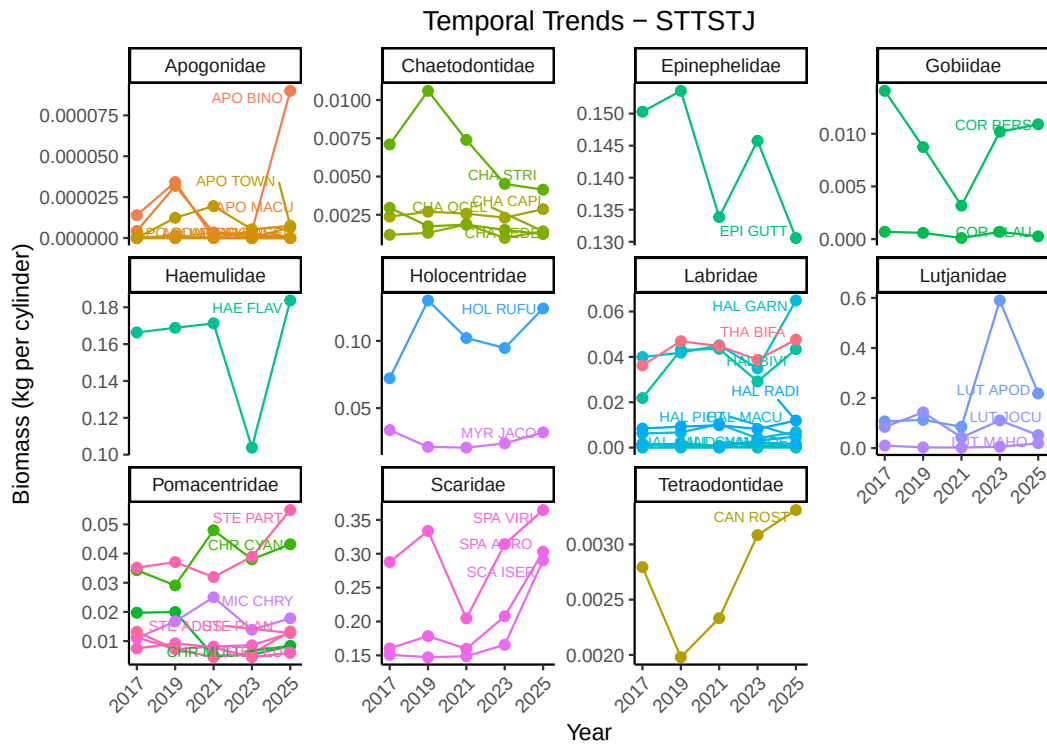
2.3 Calculate Biomass Estimates

```
## YEAR REGION STRAT PROT PRIMARY_SAMPLE_UNIT SPECIES
## 1 2017 STTSTJ AGRFDEEP 0 6018 Apogon aurolineatus
## 2 2017 STTSTJ AGRFDEEP 0 6018 Apogon binotatus
## 3 2017 STTSTJ AGRFDEEP 0 6018 Apogon maculatus
## 4 2017 STTSTJ AGRFDEEP 0 6018 Apogon phenax
## 5 2017 STTSTJ AGRFDEEP 0 6018 Apogon pseudomaculatus
```

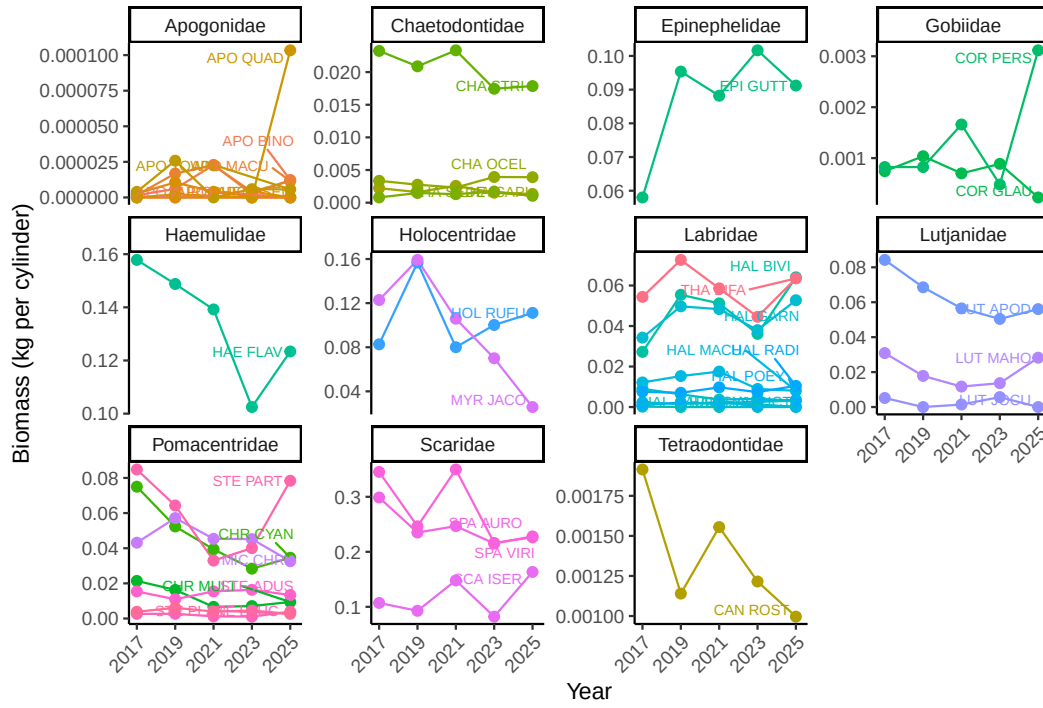
```
## 6 2017 STTSTJ AGRFDEEP 0 6018 Apogon quadrisquamatus
## SPECIES_CD group m var biomass LAT_DEGREES LON_DEGREES DEPTH
## 1 APO AURO Apogonidae 1 NA 0 18.26901 -64.7264 27
## 2 APO BINO Apogonidae 1 NA 0 18.26901 -64.7264 27
## 3 APO MACU Apogonidae 1 NA 0 18.26901 -64.7264 27
## 4 APO PHEN Apogonidae 1 NA 0 18.26901 -64.7264 27
## 5 APO PSEU Apogonidae 1 NA 0 18.26901 -64.7264 27
## 6 APO QUAD Apogonidae 1 NA 0 18.26901 -64.7264 27
```

3 Exploratory Visualization

3.1 Temporal Trends by Region

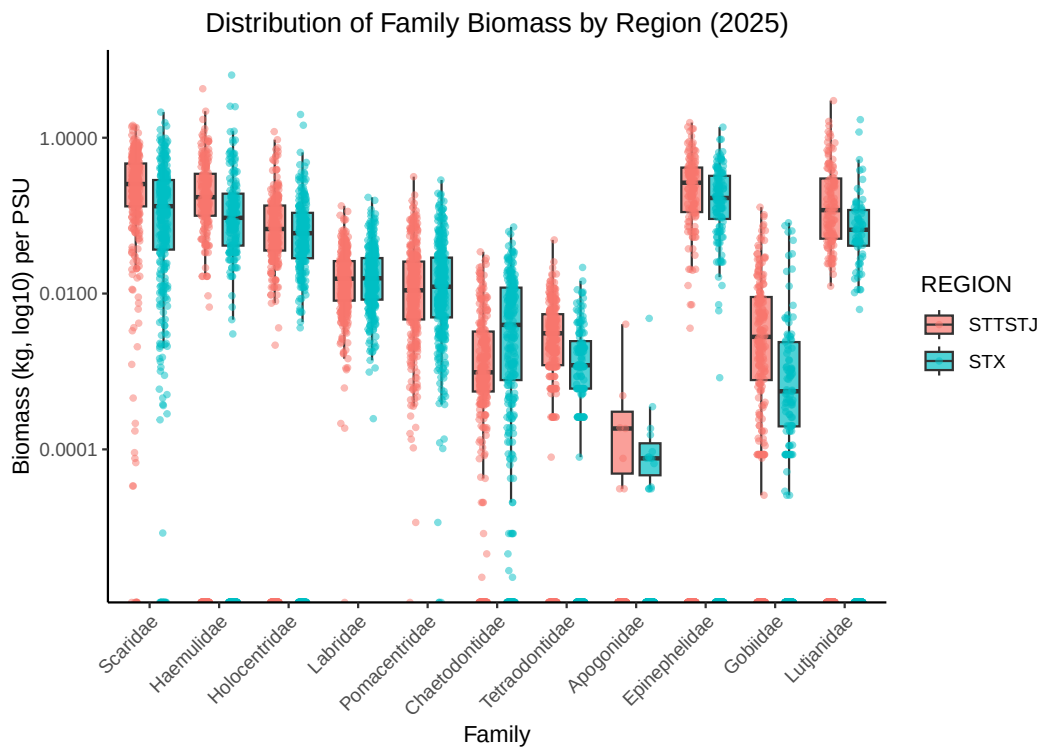
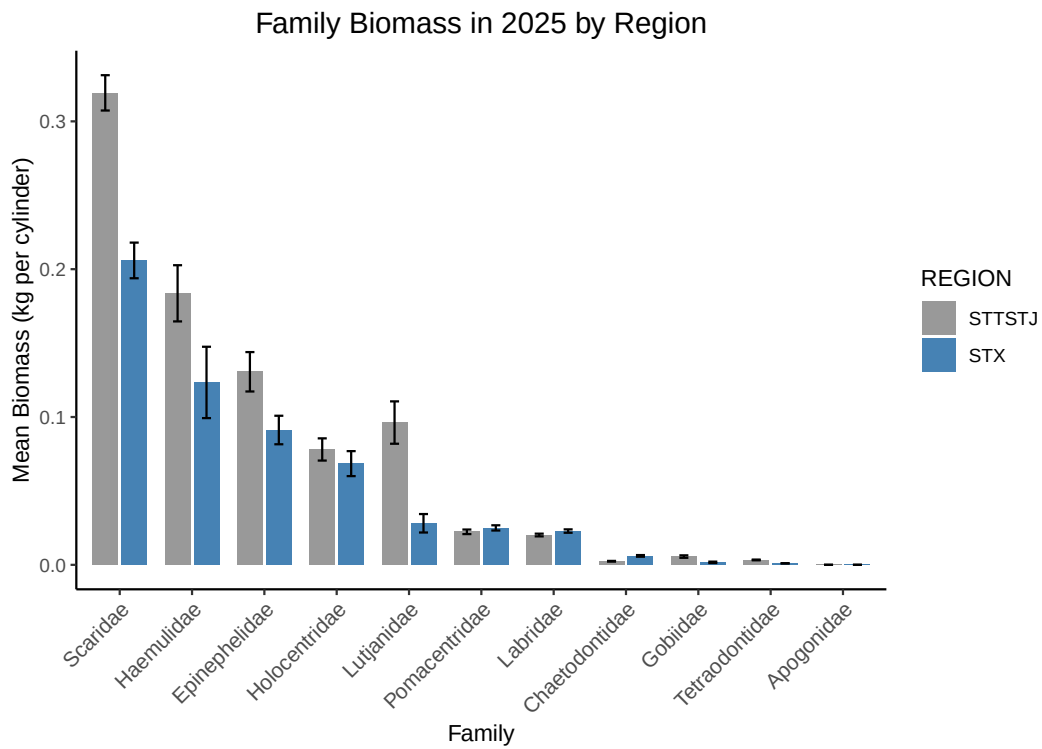


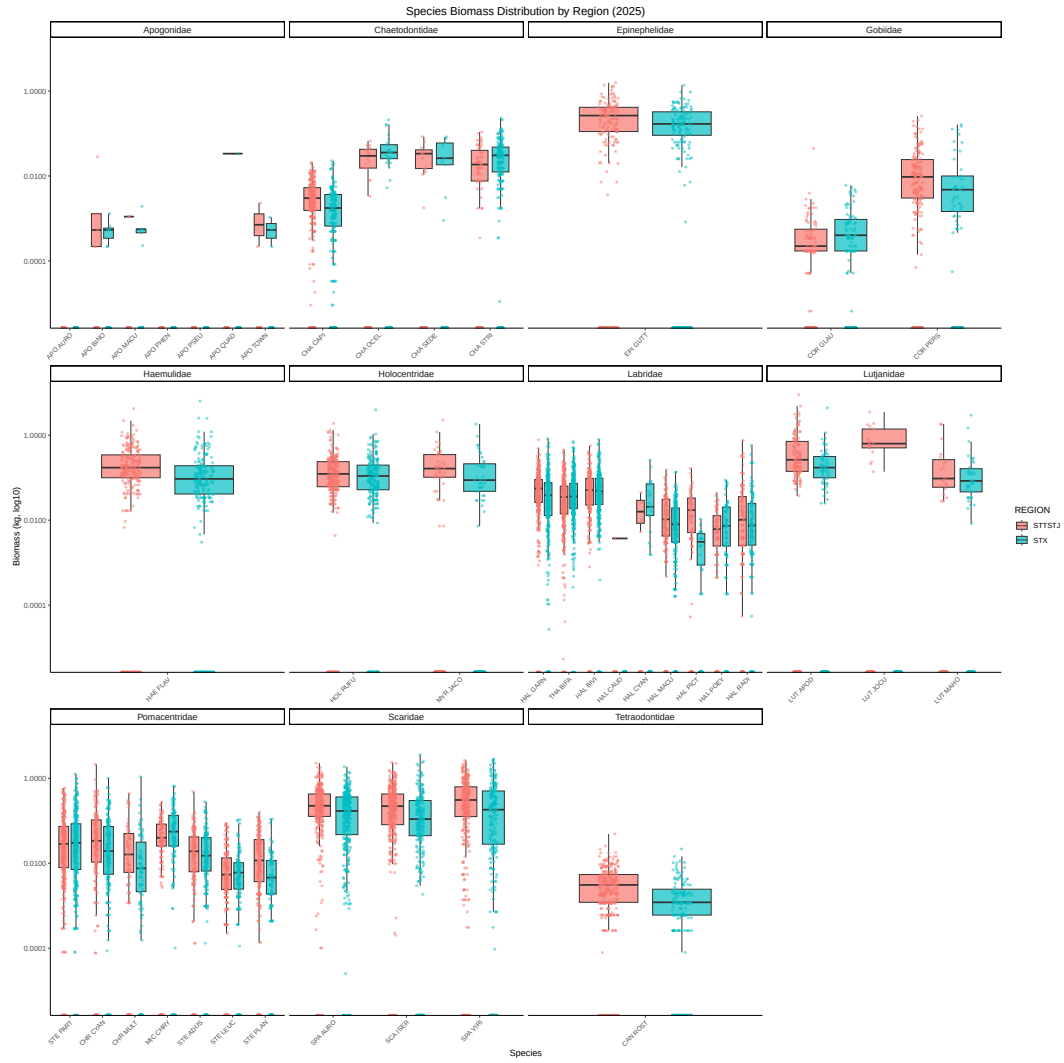
Temporal Trends – STX



4 2025 Family-Level Biomass

4.1 General plot by group





5 Statistical Analyses for Species Selection

Total observations in 2025: 26600

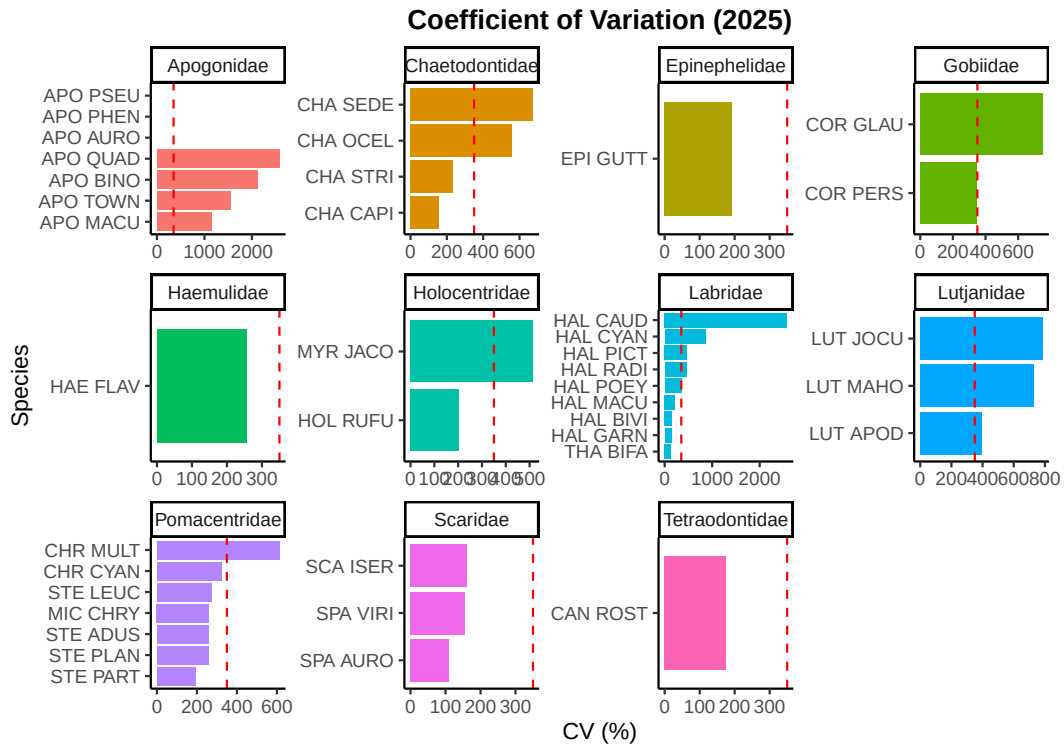
Unique PSUs: 665

Species analyzed: 40

5.1 Coefficient of Variation (CV) Analysis

The coefficient of variation ($CV = [\text{standard deviation} / \text{mean}] \times 100$) quantifies relative variability of biomass for each species across all sampling units. Lower CV values indicate more consistent biomass estimates, making species more suitable for predictive modeling. Species with $CV > 350\%$ exhibit high spatial instability that can propagate error when biomass is aggregated to the family level.

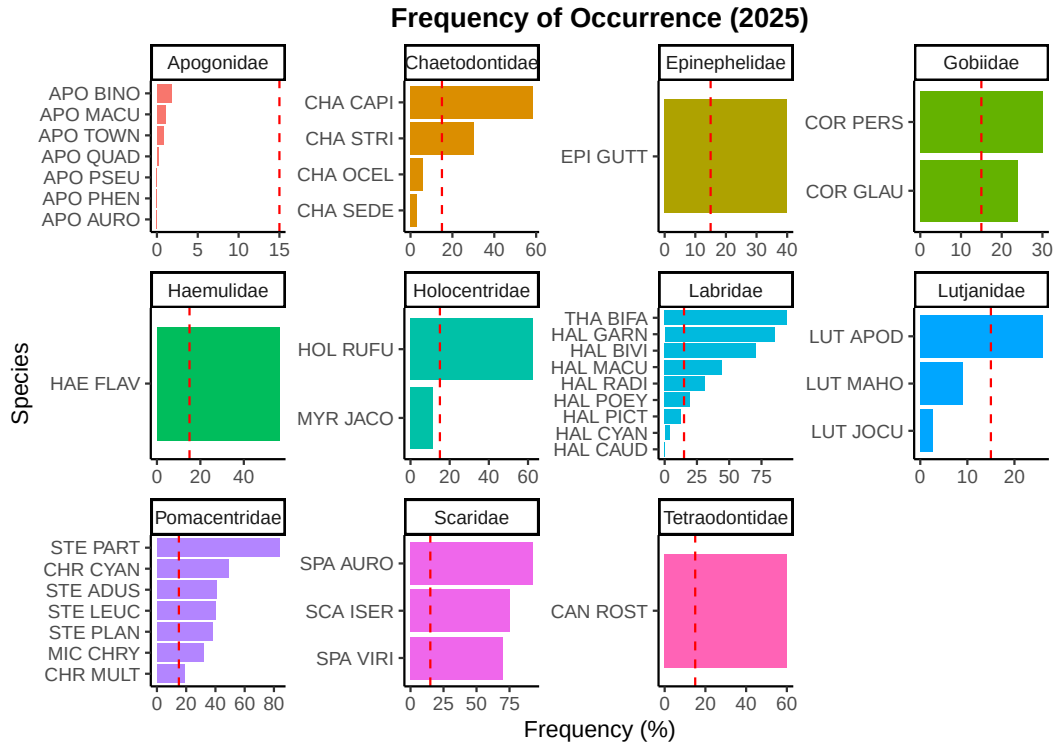
Interpretation: A species with $CV = 50\%$ shows moderate variability, while $CV > 200\%$ indicates biomass varies more than twice the mean, suggesting the species is either rare, highly patchy, or experiences boom-bust dynamics.



5.2 2. Frequency of Occurrence

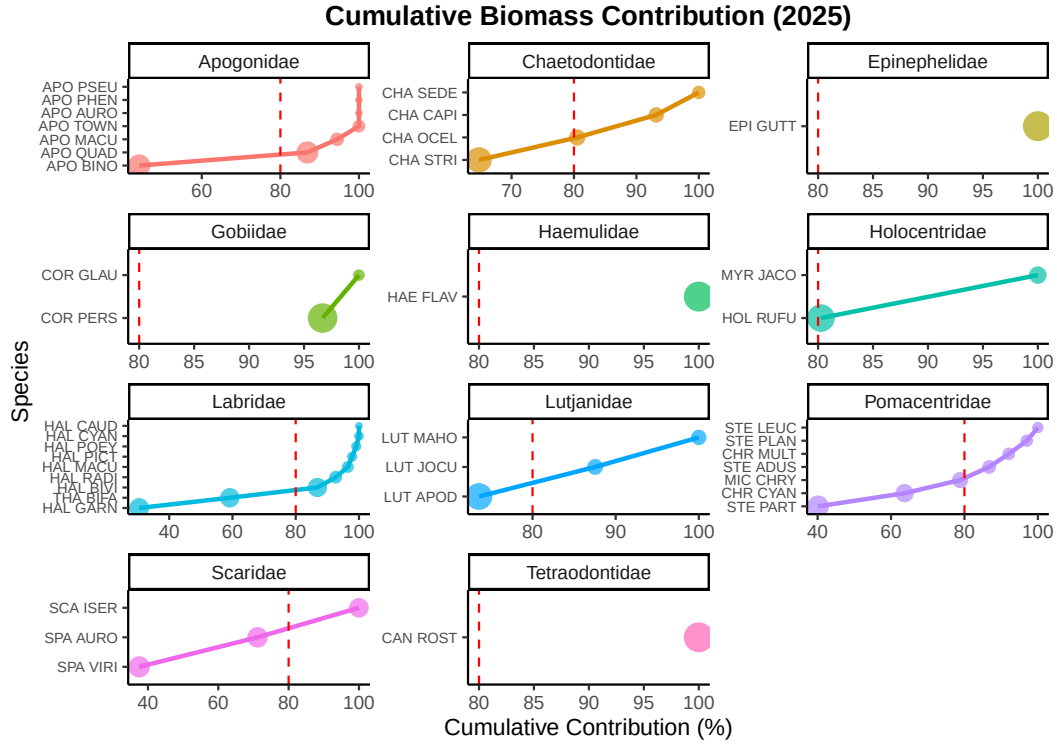
Frequency of occurrence was calculated as the percentage of sampling units where a species was present (biomass > 0). Higher frequencies indicate ubiquitous species that consistently contribute

to community biomass and are more reliably detected across the study area. Species present in <15% of samples are typically too rare to estimate biomass reliably or predict spatial patterns.



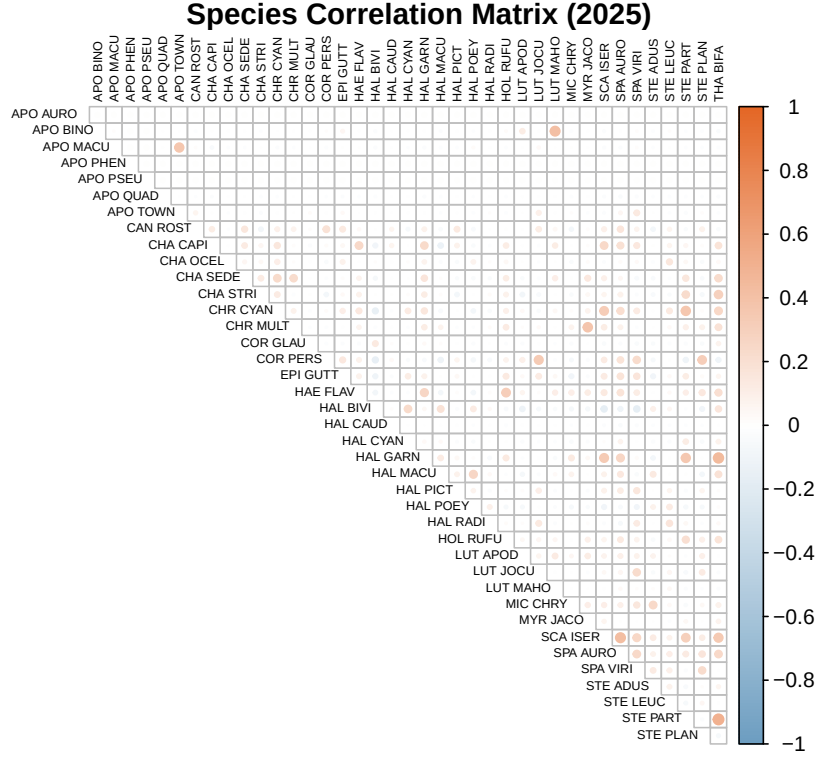
5.3 Contribution to Family Biomass

Each species' proportional contribution to family-level biomass was calculated as: $(\text{mean species biomass} / \text{mean family biomass}) \times 100$ within the 2025 dataset. This identifies dominant species that drive family-level patterns. Species contributing >30% of family biomass were considered ecologically dominant, while those contributing <10% were considered minor contributors.



5.4 Correlation Analysis

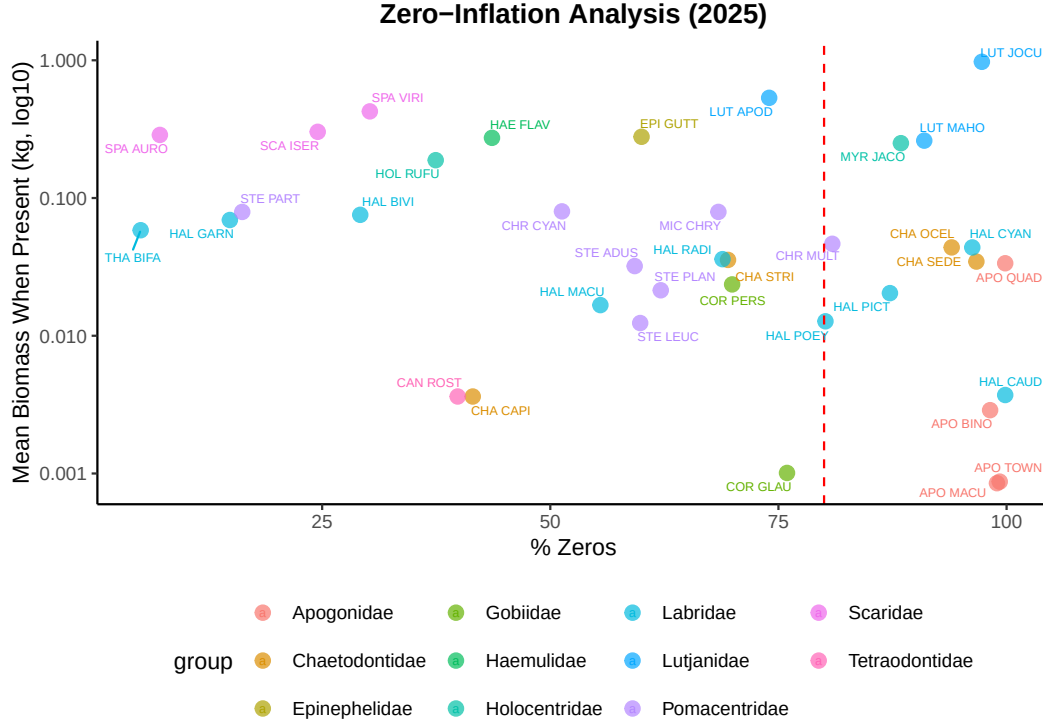
Pairwise Pearson correlations of species biomass across sampling units identified redundant species with similar spatial patterns. Species pairs with correlation coefficients >0.7 were flagged as potentially redundant, as highly correlated species provide minimal independent information but increase model complexity.



Correlation analysis revealed no species pairs with correlation coefficients exceeding $|r| = 0.7$, indicating that species exhibit largely independent spatial patterns across the study area. A total of 231 pairwise comparisons (29.6%) resulted in undefined correlations due to species having zero variance (constant or absent across all sites), further confirming the presence of hyper-rare, uninformative species in the dataset. Only two species (*Stegastes partitus* and *Thalassoma bifasciatum*) showed moderate correlation ($|r| > 0.5$) with other species, both of which were retained in the final selection due to their high quality metrics. The lack of strong inter-species correlations confirmed that removing low-quality species based on variability criteria would not result in loss of independent spatial information.

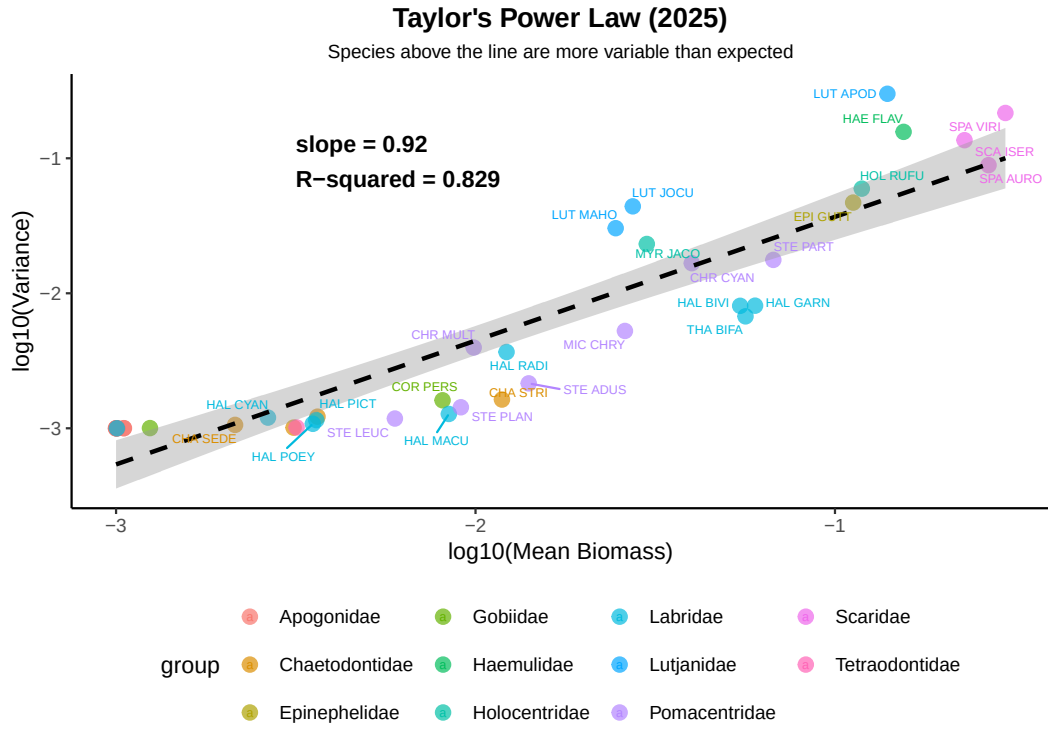
5.5 Zero-Inflation Analysis

The proportion of zero observations (absence records) was calculated for each species. High zero-inflation ($>80\%$) indicates rare or highly patchy distributions that inflate variance and complicate statistical modeling. Zero-inflated species often require specialized models and may not aggregate well with continuously present species.



5.6 Taylor's Power Law

We examined the mean-variance relationship following Taylor's Power Law: $\log(\text{variance}) = \log(a) + b \times \log(\text{mean})$. The slope parameter (b) indicates spatial aggregation patterns, where $b = 2$ suggests random distribution, $b > 2$ indicates patchy/aggregated distribution, and $b < 2$ suggests regular spacing. Species deviating strongly from this relationship represent outliers in their distributional ecology.



6 Species Selection Strategy

6.1 Integrate All Metrics

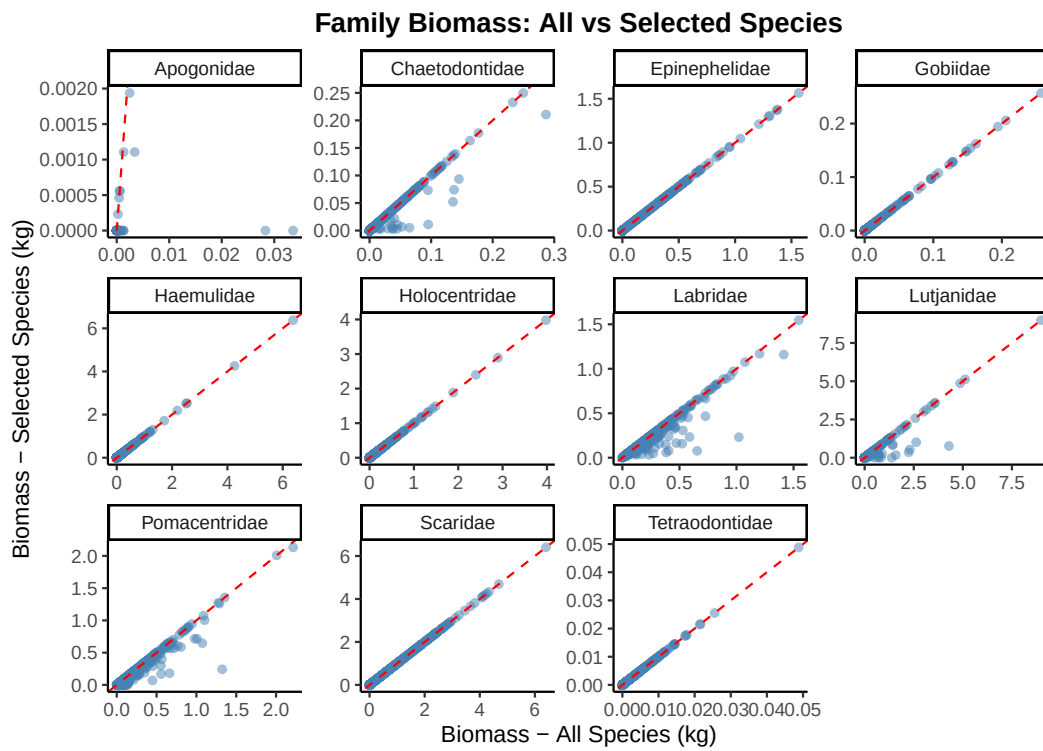
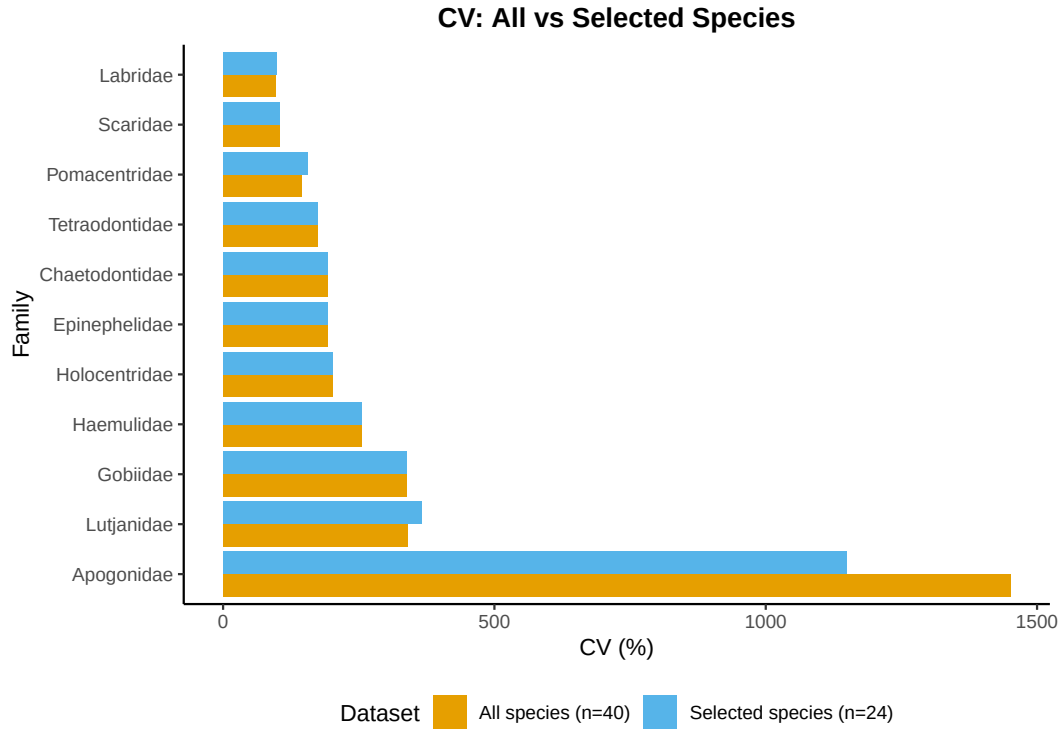
6.2 Three-Tier Selection Strategy

```
## # A tibble: 24 x 12
##   group SPECIES_CD SPECIES mean_biomass sd_biomass   cv n_obs freq_occurrence
##   <chr>  <chr>      <chr>      <dbl>      <dbl> <dbl> <int>      <dbl>
## 1 Chaet~ CHA CAPI   Chaeto~    0.00212    0.00332  157.   665      58.5
## 2 Chaet~ CHA STRI   Chaeto~    0.0108     0.0251  232.   665      30.5
## 3 Epine~ EPI GUTT   Epinep~    0.111     0.214   193.   665       40
## 4 Gobii~ COR PERS   Coryph~    0.00710    0.0247  349.   665      30.1
## 5 Haemu~ HAE FLAV   Haemul~    0.154     0.395   256.   665      56.4
## 6 Holoc~ HOL RUFU   Holoce~    0.118     0.242   206.   665      62.6
## 7 Labri~ HAL BIVI   Halich~    0.0535     0.0841  157.   665      70.8
## 8 Labri~ HAL GARN   Halich~    0.0589     0.0842  143.   665      85.1
## 9 Labri~ THA BIFA   Thalas~    0.0554     0.0758  137.   665      94.9
## 10 Pomac~ CHR CYAN   Chromi~    0.0390     0.125   322.   665      48.7
## # i 14 more rows
## # i 4 more variables: mean_contribution <dbl>, prop_zeros <dbl>,
## #   selection_score <dbl>, selection_method <chr>
```

Final Selection: 24 species across 11 families were selected:

- Tier 1 (High Quality, n=16): Mean CV = 201%, mean frequency = 62%
- Tier 2 (Dominant, n=1): *Lutjanus apodus* (74% of family biomass)
- Tier 3 (Best Available, n=7): Representatives for remaining families Two hyper-rare Apogonids (CV >2000%, <2% occurrence) were excluded.

6.3 Variance Reduction Validation



Variance Reduction Results:

Mean CV reduction: 7.5% (317% $\rightarrow$ 294%) Overall biomass retention: 96.2% Families improved: Apogonidae (21% CV reduction) Families with slight CV increases: 4 families (+0.4% to +7.8%), all retaining >86% biomass

7 Results Summary

7.1 Species Selection and Variance Reduction

Analysis of 40 candidate species using Taylor’s Power Law ($R^2 = 0.83$, slope = 0.92) revealed regular spatial spacing patterns typical of territorial reef fishes. Species exhibited high natural variability (median CV = 322%) and zero-inflation (median 70% zeros), confirming inherent patchiness.

The three-tier selection strategy identified 24 species:

Tier 1 (High Quality, n=16): Including *Sparisoma aurofrenatum* (CV=111%), *Thalassoma bifasciatum* (CV=137%), *Halichoeres garnoti* (CV=143%), and *Stegastes partitus* (CV=195%). These species provide stable biomass estimates (mean CV=201%, mean frequency=62%, mean contribution=51%).

Tier 2 (Dominant, n=1): *Lutjanus apodus* contributes 74% of Lutjanidae biomass despite moderate variability (CV=394%).

Tier 3 (Best Available, n=7): Representatives ensuring complete family coverage, including *Apogon maculatus*, *Chaetodon ocellatus*, and *Coryphopterus glaucofraenum*.

Two hyper-rare Apogonids (*A. binotatus*, *A. quadrisquamatus*) with CV >2000% and <2% occurrence were excluded.

Species selection achieved 7.5% CV reduction while retaining 96.2% of total biomass. The modest variance reduction reflects intrinsic reef fish patchiness rather than selection failure. No species pairs showed $r > 0.7$, confirming independent spatial information. The final selection balances ecological representativeness, statistical stability, and model parsimony.

7.2 Summary Table

Table 1: Summary of species selection criteria and variance reduction by family. CV = Coefficient of Variation (%). Negative CV reduction indicates slight increase. Families with CV increases retained >86% of biomass.

Family	Species (all/sel)	CV All	CV Selected	CV Reduction	Biomass Retained
Apogonidae	7/1	1451.1	1149.3	20.8	7.6
Chaetodontidae	4/3	191.9	192.7	-0.4	93.2
Epinephelidae	1/1	192.6	192.6	0.0	100.0
Gobiidae	2/2	337.6	337.6	0.0	100.0
Haemulidae	1/1	256.1	256.1	0.0	100.0
Holocentridae	2/2	202.0	202.0	0.0	100.0
Labridae	9/4	96.1	98.7	-2.7	90.7
Lutjanidae	3/2	339.4	365.9	-7.8	86.0
Pomacentridae	7/4	144.6	155.5	-7.6	86.7
Scaridae	3/3	104.6	104.6	0.0	100.0
Tetraodontidae	1/1	174.9	174.9	0.0	100.0

Family	Species (all/sel)	CV All	CV Selected	CV Reduction	Biomass Retained
Overall	40/24	317.4	293.6	7.5	96.2